

Definite integrals – 1

Darboux sums and criteria for integrability

Homework

8 сентября 2026 года

Theoretical minimum. Lower and upper Darboux sums and integrals; refinement; Darboux’s criterion in the $U - L$ form and in the oscillation form; integrability of continuous and of monotone functions; the algebra of integrable functions and the asymmetry between $\varphi \circ f$ and $f \circ \varphi$; Riemann’s covering form of the criterion.

1. Prove Riemann’s form of the criterion stated in Fact 1, and deduce that a bounded function with at most countably many points of discontinuity is integrable.
2. Prove that Thomae’s function is integrable with integral zero while the Dirichlet function is not integrable, and deduce that a pointwise limit of integrable functions need not be integrable.
3. Prove that an integrable function is integrable on every subinterval, that the integral is additive in the interval, and that f has a point of continuity in every subinterval. Do the last part with a nested interval argument, without any measure theory.
4. Prove that a uniform limit of integrable functions is integrable and that the integrals converge; then show by an example that “uniform” cannot be weakened to “pointwise and uniformly bounded” if one wants integrability of the limit.
5. Let $f \geq 0$ be integrable with $\int_a^b f = 0$. Prove that $f(x) = 0$ at every point of continuity of f , and give such an f which is not identically zero.

6. Construct a bounded f on $[0, 1]$ which is not integrable although f^2 is, and a bounded f which is not integrable although $|f|$ is. Explain which direction of the algebra of integrable functions this refutes.
7. (Iranian competition) Let $g: [0, 1] \rightarrow \mathbb{R}$ be Riemann integrable. Prove that $\int_0^1 \left(\int_x^1 g(t) dt \right) dx = \int_0^1 t g(t) dt$, first by a partition argument and then by changing the order of integration.
8. (MIPT) Let f and g be bounded 2π -periodic functions and let α, β be real numbers. Suppose that $f(t) - f(t + \alpha) + g(t) - g(t + \beta)$ is Riemann integrable on every finite interval. Find all possible values of its integral over $[0, 2\pi]$.
9. Prove that for a bounded f the upper integral equals $\inf \int_a^b h$ over all step functions $h \geq f$, and the lower integral equals the corresponding supremum. Deduce Darboux's criterion once more from this description.
10. Let f be integrable on $[a, b]$. Prove that $\lim_{h \rightarrow 0} \int_a^{b-h} |f(x+h) - f(x)| dx = 0$ when f is continuous, and show that for a general integrable f the same limit still holds.